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LAB BOOK UNIT 3

1) Write a simple procedure without any parameter that shows user defined message on the screen. Call the procedure using separate PL/SQL block and on the command line.

```
CREATE OR REPLACE PROCEDURE show_message
IS
BEGIN
    DBMS_OUTPUT.PUT_LINE('Hello! This is a user-defined message.');
```

—Call using separate PL/SQL block

```
END;
/
```

—Call on command line

```
EXEC show_message;
```


2) Write a simple procedure that increases the basic salary of employees for the given department no by percentage inputted by the user using IN parameter.

```
CREATE OR REPLACE PROCEDURE increase_salary(
    p_deptno IN emp.deptno%TYPE,
    p_percent IN NUMBER
)
IS
BEGIN
    UPDATE emp
    SET sal = sal + (sal * p_percent / 100)
    WHERE deptno = p_deptno;
```

```

    DBMS_OUTPUT.PUT_LINE('Salary updated successfully. ');
END;
/

```

—Call

```

BEGIN
    increase_salary(10, 5);
END;
/

```

```

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-----

```

3) Write a procedure that searches whether the given employee id is present or not in the table. If employee is found then show its name otherwise raise appropriate error message (Use both IN and OUT mode variables) and also write a PL/SQL block to call the procedure.

```

CREATE OR REPLACE PROCEDURE find_employee(
    p_empid IN emp.empno%TYPE,
    p_ename OUT emp.ename%TYPE
)
IS
BEGIN
    SELECT ename
    INTO p_ename
    FROM emp
    WHERE empno = p_empid;

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        RAISE_APPLICATION_ERROR(-20001, 'Employee not found. ');
END;
/

```

—Call

```

DECLARE
    v_name emp.ename%TYPE;
BEGIN
    find_employee(7369, v_name);
    DBMS_OUTPUT.PUT_LINE('Employee Name: ' || v_name);
END;
/

```

```

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```

4) Write a function that returns the square of the given number. Execute the function using separate PL/SQL block and on the command line.

```
CREATE OR REPLACE FUNCTION find_square(  
    p_num IN NUMBER  
)  
RETURN NUMBER  
IS  
BEGIN  
    RETURN p_num * p_num;  
END;  
/
```

—Execute using separate PL/SQL block.

```
DECLARE  
    v_result NUMBER;  
BEGIN  
    v_result := find_square(5);  
    DBMS_OUTPUT.PUT_LINE('Square is: ' || v_result);  
END;  
/
```

—Execute on command line

```
SELECT find_square(5) FROM dual;
```

```
-----  
-----
```

5) Write a function that returns balance for given account number. (Create ACCOUNT table with ACNO, CNAME, BNAME, BALANCE columns using appropriate datatypes)

—create table

```
CREATE TABLE account (  
    acno NUMBER(5) PRIMARY KEY,  
    cname VARCHAR2(30),  
    bname VARCHAR2(30),  
    balance NUMBER(10,2)  
);
```

—Function

```
CREATE OR REPLACE FUNCTION get_balance(  
    p_acno IN account.acno%TYPE  
)  
RETURN NUMBER  
IS  
    v_balance account.balance%TYPE;  
BEGIN  
    SELECT balance  
    INTO v_balance  
    FROM account  
    WHERE acno = p_acno;  
    RETURN v_balance;
```

EXCEPTION

WHEN NO_DATA_FOUND THEN

RAISE_APPLICATION_ERROR(-20002, 'Account not found.');

END;

/

—Execution

SELECT get_balance(101) FROM dual;

